

Module 5: Future Nonequilibrium Electron-Phonon and Photon Physics

Photon absorption, carrier excitation, relaxation, recombination, and pair breaking

Physics note for FEM_BallisticDiffusion_PINN

Purpose. This note documents physics that should be added after the first FEM and PINN framework is stable. No code is required in this module for the first build. The goal is to create a clear roadmap for future nonequilibrium physics: photon absorption, hot-electron generation, electron-phonon relaxation, high-energy phonon creation, Cooper-pair breaking, quasiparticle recombination, and phonon bottlenecks.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
$h\nu$	Photon energy	Energy of an absorbed photon of frequency ν .
Δ	Superconducting energy gap	Minimum quasiparticle excitation energy scale.
2Δ	Pair-breaking threshold	Minimum energy required to break a Cooper pair into two quasiparticles.
N_{qp}	Quasiparticle number	Total number of quasiparticles in a modeled region.
N_{Ω}	Pair-breaking phonon number	Number of high-energy phonons capable of breaking Cooper pairs.
τ_R	Recombination time	Characteristic time for quasiparticle recombination.
τ_B	Pair-breaking time	Characteristic time for high-energy phonons to break pairs.
τ_{esc}	Phonon escape time	Time for high-energy phonons to leave the superconducting film or escape into substrate/sink.
Σ	Electron-phonon coupling coefficient	Material parameter in reduced electron-phonon heat-transfer laws.
T_e	Electron temperature	Effective temperature of electron/quasiparticle subsystem.
T_{ph}	Phonon temperature	Effective temperature of phonon/lattice subsystem.
RT	Rothwarf-Taylor	Coupled rate-equation model for quasiparticles and pair-breaking phonons.

1 Why this is future work

Modules 1–4 create a usable reduced transport framework. Module 5 is more microscopic and more nonequilibrium. It involves energy-resolved populations, pair breaking, phonon escape, and electron-phonon relaxation. These processes are essential for a mature model, but adding them before the geometry/FEM framework works would make the project too complicated too early.

The future extension should answer questions such as:

1. Where is a photon absorbed?
2. How much of the photon energy creates quasiparticles?
3. How much becomes high-energy phonons?
4. Do those phonons escape, thermalize, or break more Cooper pairs?
5. How do traps and heat sinks alter these pathways?

2 Photon absorption and pair-breaking threshold

A photon with energy

$$E_\gamma = h\nu \quad (2.1)$$

can break a Cooper pair if it can supply at least

$$E_\gamma \geq 2\Delta. \quad (2.2)$$

The factor of two appears because a Cooper pair must be broken into two quasiparticle excitations, each costing at least roughly Δ in energy.

Not every absorbed photon produces the same number of quasiparticles. Some energy may remain in hot electrons, some may go into high-energy phonons, and some may escape before generating additional quasiparticles. A reduced source term can be written as

$$S_{\text{qp}}(\mathbf{r}, t) = \eta_{\text{qp}} \frac{Q_{\text{abs}}(\mathbf{r}, t)}{\Delta}, \quad (2.3)$$

where Q_{abs} is absorbed power density and η_{qp} is an efficiency factor. The factor Δ converts energy rate density into approximate quasiparticle generation rate density.

3 Electron-phonon relaxation

After energy is deposited in a metal film, the electronic subsystem and phonon subsystem may not immediately share one temperature. A reduced two-temperature model is

$$C_e(T_e) \frac{\partial T_e}{\partial t} = \nabla \cdot (k_e \nabla T_e) - G_{e-ph}(T_e, T_{ph}) + Q_{\text{abs}}, \quad (3.1)$$

$$C_{ph}(T_{ph}) \frac{\partial T_{ph}}{\partial t} = \nabla \cdot (k_{ph} \nabla T_{ph}) + G_{e-ph}(T_e, T_{ph}) - G_{\text{esc}}(T_{ph}, T_{\text{sub}}). \quad (3.2)$$

At low temperature, a common normal-metal electron-phonon energy-transfer form is

$$G_{e-ph} = \Sigma (T_e^5 - T_{ph}^5). \quad (3.3)$$

In a superconductor, the gapped density of states modifies these laws. Therefore this form should be treated as a conceptual bridge, not a final superconducting microscopic model.

4 Rothwarf-Taylor rate equations

A classic reduced model for nonequilibrium quasiparticles and high-energy phonons is the Rothwarf-Taylor system. In a spatially lumped form,

$$\frac{dN_{\text{qp}}}{dt} = -RN_{\text{qp}}^2 + 2\beta N_{\Omega} + G_{\text{qp}}, \quad (4.1)$$

$$\frac{dN_{\Omega}}{dt} = \frac{1}{2}RN_{\text{qp}}^2 - \beta N_{\Omega} - \frac{N_{\Omega}}{\tau_{\text{esc}}} + G_{\Omega}. \quad (4.2)$$

Here N_{Ω} is the number of high-energy phonons capable of breaking Cooper pairs, R is a recombination coefficient, β is a pair-breaking rate by high-energy phonons, τ_{esc} is the phonon escape time, and G_{qp} and G_{Ω} are generation terms.

The physical feedback loop is important. Recombination removes two quasiparticles, but it emits a phonon with energy near 2Δ . If that phonon remains in the film, it can break another Cooper pair. This produces a phonon bottleneck. A good heat sink or substrate escape pathway can reduce this bottleneck by removing high-energy phonons.

5 Spatial extension

A spatially resolved future model could couple quasiparticle diffusion and high-energy phonon diffusion:

$$\frac{\partial n_{\text{qp}}}{\partial t} = \nabla \cdot (D_{\text{qp}} \nabla n_{\text{qp}}) - Rn_{\text{qp}}^2 + 2\beta n_{\Omega} + S_{\text{qp}} - \Gamma_{\text{trap}} n_{\text{qp}}, \quad (5.1)$$

$$\frac{\partial n_{\Omega}}{\partial t} = \nabla \cdot (D_{\Omega} \nabla n_{\Omega}) + \frac{1}{2}Rn_{\text{qp}}^2 - \beta n_{\Omega} - \frac{n_{\Omega}}{\tau_{\text{esc}}} + S_{\Omega}. \quad (5.2)$$

This is a natural Module 5 future upgrade because it connects directly to the geometry already built in Module 3.

6 Relaxation and recombination cascade

A realistic absorbed-energy event may pass through the following stages:

1. photon or energetic phonon deposits energy in a film, substrate, or junction region,
2. high-energy electronic excitations are created,

3. electron-electron scattering redistributes energy inside the electronic subsystem,
4. electron-phonon scattering transfers energy to phonons,
5. phonons with energy above 2Δ can break Cooper pairs,
6. quasiparticles diffuse and recombine,
7. recombination phonons either escape, down-convert, or break pairs again,
8. traps and heat sinks modify the escape and removal rates.

7 How Module 5 connects back to earlier modules

The future nonequilibrium model should not replace Modules 1–4. It should provide better source and coupling terms:

$$S_{\text{qp}}(\mathbf{r}, t), \quad Q_{\text{abs}}(\mathbf{r}, t), \quad \Gamma_{\text{trap}}(\mathbf{r}), \quad \tau_{\text{esc}}(\mathbf{r}), \quad \eta_{\text{qp}}(\mathbf{r}). \quad (7.1)$$

These improved terms can then be inserted into the FEM and PINN framework already established.

8 First future-work milestones

The recommended order for future work is:

1. add a spatially localized photon absorption source term,
2. add an energy-to-quasiparticle conversion efficiency,
3. add a lumped Rothwarf-Taylor model coupled to the junction region,
4. extend the lumped model to spatially resolved n_{qp} and n_{Ω} ,
5. couple phonon escape to substrate and heat-sink geometry,
6. train a PINN surrogate for the expanded nonequilibrium model.

9 References

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